CHANNING BLOEDEL

Brookings, SD 57006, (605) 591-2237, channingbloedel@gmail.com

SKILLS

Technical Skills: CAD (SolidWorks, Autodesk Inventor, Creo), Analysis (ANSYS, StarCCM+, ADAMS), MATLAB, Autodesk ReCap, Prototyping, Basic Machining Proficiency, Basic Maintenance & Repair, Microsoft Office, Capable

EDUCATION

South Dakota State University, Brookings, SD May 2028

- Pursuing Mechanical Engineering Bachelor of Science
- GPA 3.3/4.0
- Relevant coursework upon Summer 2027: Thermodynamics II, Heat Transfer, Automatic Controls, Aerodynamics, Design of Machine Elements, Engineering Materials, Engineering Design Methods, Advanced Engineering Math

WORK EXPERIENCE

Mechanical Design Intern – Aerofly October 2025 – October 2026

- Built digital twin models of 15+ electrical substations in Autodesk Inventor from Autodesk ReCap scan data
- Worked closely with coworkers to align designs across substation systems, and learning new workflows
- Led quality control efforts and created systematic processes to streamline workflow

Lab Services Reliability Mechanical Engineering Student - Daktronics April 2025 – September 2025

- Ran hands-on reliability tests on power supply systems used in LED displays
- Conducted physical construction analysis to verify component integrity and design alignment
- Executed failure analysis investigations to find root causes of power system malfunctions
- Documented failure analysis findings and collaborated with engineering teams for product improvements
- Designed parts in Creo to advance development of new testing methods

District Technology Assistant - Lead-Deadwood School May 2024 – August 2024

- Processed the surplus of over 200 pieces of hardware
- Imaged and prepared technology for ~450 students

Freelance Maintenance - Miller Apartments September 2016 – July 2024

- Ran hands-on repair and renovation jobs for a residential property for 7+ years, estimating costs on 50+ jobs

ACTIVITIES

- **NASA RASC-AL 2027 Project "TBD" – Team Leader** August 2026 – Present
  - Systems engineering competition to design a lunar transit pathway construction mission
  - Leading a team of 15 to start, grow, and learn new aerospace skills together
  - Leading a lunar regolith simulant sintering and compressive strength **experiment**
- **NASA RASC-AL 2026 Project "SELENE" – Design Lead** March 2026 – June 2026
  - Systems engineering competition to design a sample storage mission to the moon
  - Designed a refrigerated and ambient storage container to hold lunar surface core samples
  - Selected as **Best in Theme (Finalist)**
- **NASA RASC-AL 2026 Project "PROMETHEUS" - Team Leader** September 2025 – March 2026
  - Systems engineering competition to design an architecture to power the moon
- **NASA RASC-AL 2025 Project "NOVA" - Mobility Sub-Team Leader** September 2024 – June 2025
  - Systems engineering competition to design a lunar maintenance and servicing rover
  - Lead collaboration within a small team to design an effective lunar rover mobility system
  - Designed and FEA'd rover mobility system using SolidWorks, ANSYS, and ADAMS View, and prototyped
  - Awarded **Best Prototype (Finalist)**
- **NASA X-Hab Project** April 2025 – December 2025
  - Assisted on proposal team to obtain this project for South Dakota State University's senior design class
  - Stepped in hands-on wherever needed, completing time-sensitive objectives
- **Oasis Leadership Team** March 2025 – Present